

Tuning the model

The next step is model tuning or hyperparameter optimisation. This is when the model parameters are adjusted to enhance its performance. Hyperparameters are parameters that are set before training the model and control aspects like learning rate, regularisation strength, or the number of trees in a random forest.

```
from sklearn.model_selection
import GridSearchCV
Defining parameter grid
param_grid = {'C': [0.1, 1, 10], 'solver': ['liblinear',
'saga']}
Grid search
grid_search = GridSearchCV(LogisticRegression(), param_grid,
cv=5) grid_search.fit(X_train_scaled, y_train)
Best parameters
print(grid_search.best_params_)
```

Model deployment

After successfully training, evaluating, and tuning your model, the next step is deployment. During this process, the model is made available to end users or systems for real-time

predictions. Common approaches to deployment include:

```
import joblib
Saving the model
joblib.dump(model, 'logistic_model.pkl')
Loading the model
model = joblib.load('logistic_model.pkl')
Making predictions
y_pred = model.predict(X_test_scaled)
```

Python has paved its way as a language for machine learning due to its readability and libraries like scikit-learn, TensorFlow, and PyTorch. Its algorithms help developers and data scientists to build effective and efficient machine learning models. As continuous improvements are made in Python's algorithms and as computational power enhances, it will continue to partner with machine learning to fuel innovations. **END** 

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